PORTABLE DATA CARTRIDGE WITH OPTICAL INTERFACE

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March 8, 2026

Title: PORTABLE DATA CARTRIDGE WITH OPTICAL INTERFACE

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Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, filed on or about March 6, 2026, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A portable, self-contained, high-capacity data storage cartridge designed for the rapid physical transport of large datasets. The apparatus comprises a ruggedized housing containing a high-density, non-volatile storage medium, an integrated power source (e.g., a rechargeable battery), and at least one massively parallel optical interface for data ingest and offload. The optical interface, consistent with the technology of the parent application, allows the cartridge to be loaded or unloaded with petabyte-scale data in minutes, rather than the hours or days required with conventional interfaces. The cartridge is the foundational hardware for a new type of physical data logistics network, making the physical transfer of data a viable and often faster alternative to network transfer for bulk datasets.

Background of the Invention

As datasets in fields like AI, genomics, and national security grow into the petabyte and exabyte scale, transferring them over conventional telecommunication networks has become increasingly impractical due to bandwidth limitations and security concerns. The alternative, physically transporting storage media (a practice often called "sneakernet"), is limited by the slow I/O speeds of traditional portable hard drives or storage arrays. The time taken to load and unload the data at either end often negates the benefits of physical transport. There is a need for a portable storage device that combines high storage capacity with an ultra-high-bandwidth interface to make physical data logistics truly efficient.

Summary of the Invention

The present invention is a portable data cartridge with a high-bandwidth optical interface, colloquially referred to as a "data brick," that is specifically designed to solve the I/O bottleneck in physical data transport. The cartridge is a self-contained unit comprising three key components within a durable, shock-resistant housing: a high-capacity non-volatile storage medium, an internal power source, and a pair of massively parallel optical interfaces (one for input, one for output).

The optical interfaces are consistent with the CVBNI architecture described in the parent application, No. 19/559,705. They enable the cartridge to ingest or offload its entire data payload at multi-terabit-per-second speeds. This reduces the "loading dock" time for a petabyte of data from many hours to mere minutes, making the physical transport of the cartridge via courier, drone, or vehicle an extremely fast and efficient method for bulk data transfer. The cartridge is the key enabling apparatus for a new ecosystem of secure physical data logistics.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is an isometric view of the Portable Data Cartridge with Optical Interface, showing its ruggedized housing and optical I/O ports.

FIG. 2 is a block diagram illustrating the internal components of the data cartridge.

Detailed Description of the Invention

Referring to **FIG. 1**, the apparatus is a portable data cartridge (100). It is enclosed in a ruggedized, shock-absorbent housing (101) suitable for physical transport. The housing includes at least two massively parallel optical ports: an input port (102) for receiving data and an output port (103) for transmitting data. These ports are protected by a retractable, sealed cover (104) to prevent contamination of the optical surfaces during transit. The housing may also include status indicators (105) for power and data activity.

FIG. 2 is a block diagram showing the internal components of the cartridge (100). The core components are a high-capacity, non-volatile storage medium (201), a storage controller (202), and an integrated power source (203), such as a high-capacity rechargeable battery. The storage controller (202) manages the flow of data to and from the optical interfaces. The input optical port (102) is coupled with a photon sensor array (204), while the output optical port (103) is coupled with a photon emitter array (205). This bidirectional design with separate input and output arrays allows for maximum data transfer speed and efficiency.

Claims

What is claimed is:

1. A portable data storage apparatus, comprising:
    - a ruggedized housing designed for physical transport;
    - a non-volatile data storage medium contained within the housing;
    - a self-contained power source contained within the housing; and
    - at least one massively parallel optical interface integrated into the housing for data input and output.

2. The apparatus of claim 1, comprising a first massively parallel optical interface for data input and a second, separate massively parallel optical interface for data output.

3. The apparatus of claim 2, wherein the first interface comprises a two-dimensional array of photon sensors and the second interface comprises a two-dimensional array of photon emitters.

4. The apparatus of claim 1, wherein the apparatus is configured to transfer data via the optical interface at a rate exceeding 1 terabit per second.

5. The apparatus of claim 1, further comprising a protective, sealed cover for the optical interface.

Drawings

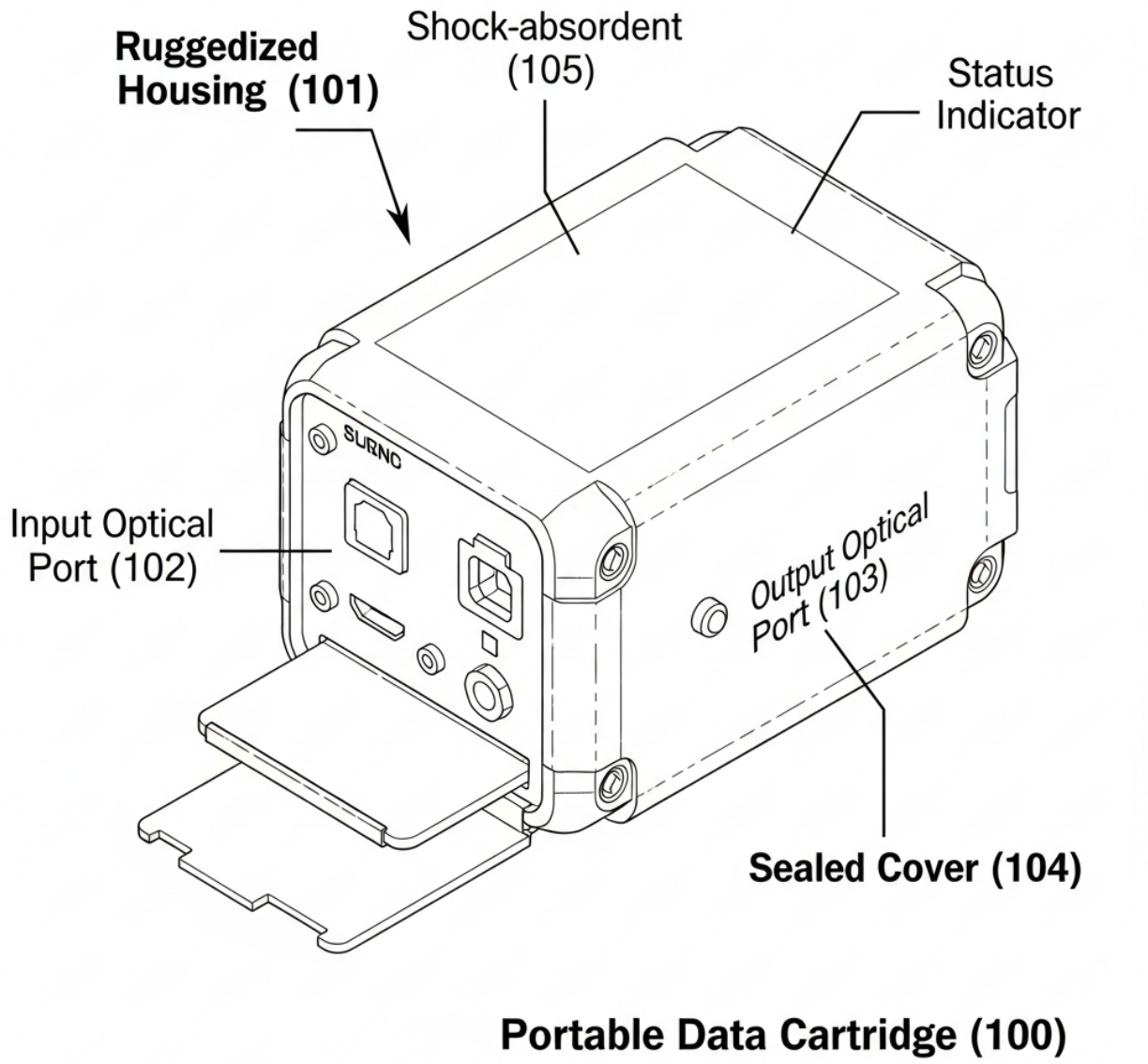


Figure 1: An isometric view of the Portable Data Cartridge with Optical Interface, showing its ruggedized housing and optical I/O ports.

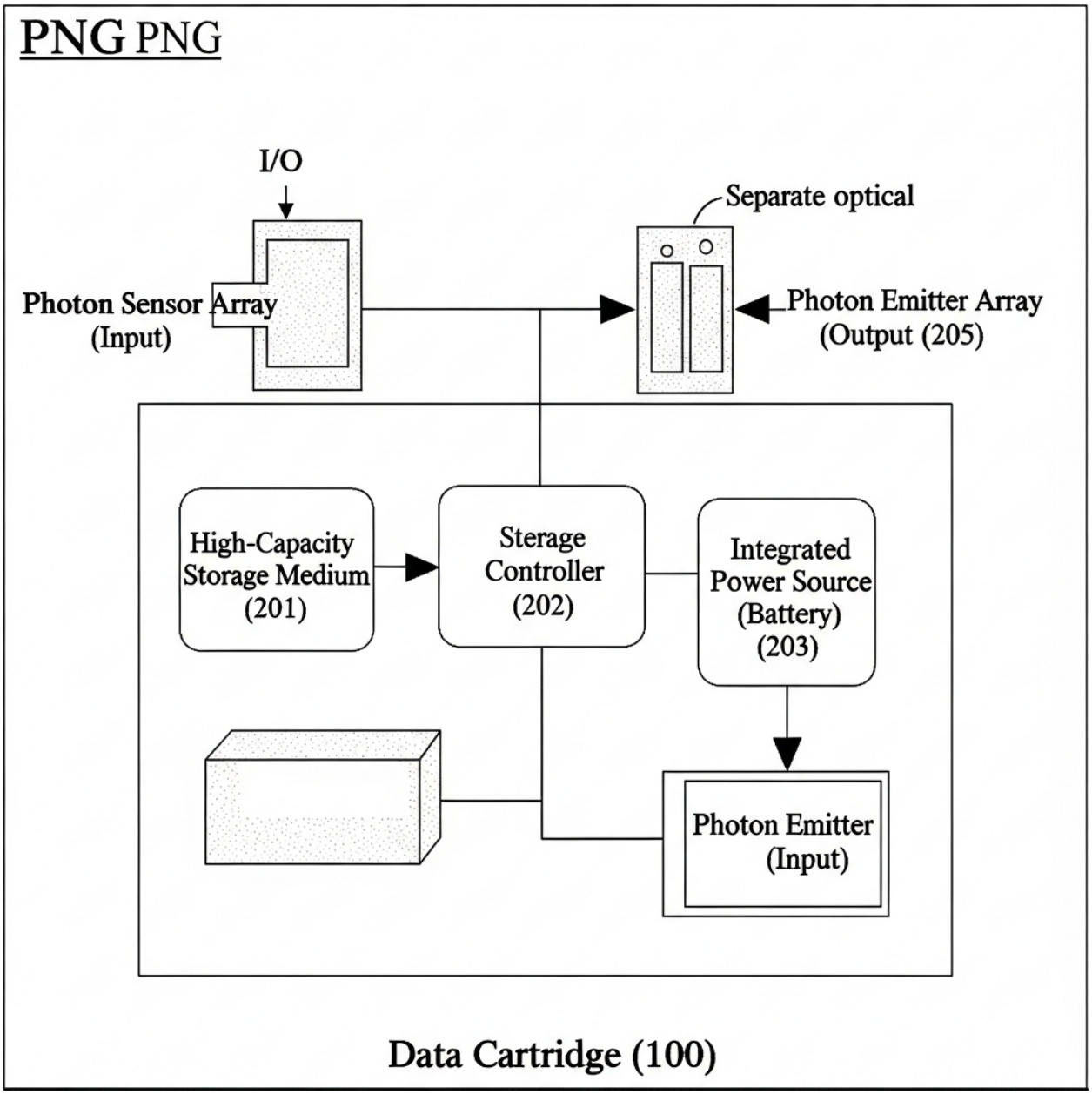


Figure 2: A block diagram illustrating the internal components of the data cartridge.